

CE902 2021 Worked Solutions (Section B Only)

Professional Practice and Research Methodology — explanation-first solution booklet

Source note: the uploaded exam file explicitly states that Section A questions are not available. Accordingly, this booklet solves all visible questions in Section B only (Questions 18–21), with formulas, working, reasoning, and structured written responses.

Question 18 — Project appraisal using discounting and NPV

The three projects have different timings of cash inflow and different starting costs. A fair comparison therefore requires discounting future cash receipts back to today and then subtracting the initial capital cost to obtain net present value (NPV).

Present value formula: $PV = X / (1 + r)^t$, where X is the cash flow in year t and $r = 0.08$.

Illustrative discounting timeline for Project A (£000):



Each end-of-year cash inflow is converted to present value using $PV = X / (1+r)^t$ with $r = 0.08$.

18(a) Discount factors for years 1-5

Year	Discount factor = $1/(1.08)^t$	Rounded value
1	$1 / (1.08)^1$	0.925926
2	$1 / (1.08)^2$	0.857339
3	$1 / (1.08)^3$	0.793832
4	$1 / (1.08)^4$	0.735030
5	$1 / (1.08)^5$	0.680583

18(b) Discounted cash inflows for each project

Each yearly inflow is multiplied by the corresponding discount factor. For instance, Project A in Year 3 gives $PV = 125 \times 0.793832 = 99.229$ (£000, approximately).

Year	Discount factor	Project A PV (£000)	Project B PV (£000)	Project C PV (£000)
1	0.925926	69.444	92.593	46.296
2	0.857339	107.167	171.468	64.300
3	0.793832	99.229	238.150	198.458
4	0.735030	73.503	220.509	220.509
5	0.680583	51.044	102.087	136.117
Total discounted inflow		400.388	824.806	665.680

18(c) NPV comparison and decision

Project	Total discounted inflow (£000)	Initial capital cost (£000)	NPV = inflow – cost (£000)
A	400.388	300.000	100.388
B	824.806	500.000	324.806
C	665.680	300.000	365.680

All three NPVs are positive, which means each project beats the 8% benchmark. However, Project C has the largest NPV (approximately £365.680k) and is therefore the most financially attractive option. Project B earns more discounted cash in total than A, but its larger initial cost leaves it below Project C.

Question 19 – Business structure and gearing

19(a) Five differences between a limited company and a sole trader

Aspect	Limited company	Sole trader
Legal status	Separate legal entity from the owner(s)	No separate legal identity; owner and business are legally the same
Liability	Owners/shareholders have limited liability	Owner has unlimited personal liability
Ownership	Owned by shareholders	Owned by one individual
Taxation	Usually pays corporation tax; owners may also pay tax on salary/dividends	Owner usually pays income tax on business profits
Continuity	Business can continue despite ownership change	Business is closely tied to the owner

19(b) Meaning of gearing (leverage) and examples

Gearing measures the extent to which a company is financed by borrowed money rather than by owners' funds. A highly geared company therefore has a larger debt burden relative to equity or assets.

Common gearing ratio	Illustrative form
Debt-to-equity ratio	Total debt / Shareholders' equity
Debt ratio	Total debt / Total assets
Interest cover ratio	Operating profit / Interest payable

19(c) Why high gearing is riskier

A highly geared company must meet larger fixed interest and repayment commitments. If profits or cash flow weaken, those obligations remain, so the chance of financial stress or default rises. High gearing also reduces flexibility because more operating cash is already committed to servicing debt.

Question 20 – Referencing, summarising, and plagiarism

20(a) Is a reference required for 'Tim Berners-Lee invented the World Wide Web'?

Usually no reference is required if the statement is being used as common knowledge. It is a widely known historical fact rather than a specialised claim. A reference would become more appropriate if the assignment were making a detailed historical argument or relying on a specific interpretation.

20(b) What belongs in a summary and what does not?

Item	Include?	Reason
A. Key points from the article	Yes	A summary should communicate the main ideas.

Item	Include?	Reason
B. The author's viewpoint	Yes	A summary should fairly reflect the source author's position.
C. Your viewpoint on the topic	No	This belongs in your own discussion, not in a summary.
D. In-text referencing notes such as page numbers for direct quotes	No	These are citation mechanics rather than summary content.
E. Your personal opinion about the article	No	Personal evaluation belongs in critique, not in the summary itself.

Academically, the strongest answer is that only A and B belong in the content of the summary itself. The exam wording asks for three to include, but D is better seen as referencing apparatus rather than summary content.

20(c) Does the student extract amount to plagiarism?

Yes. The passage rewrites the source very closely, keeping the same sequence of ideas, specialist terms, and overall conceptual structure, but it gives no citation. Changing some words is not enough. This is still unattributed use of another author's ideas and phrasing, so it amounts to plagiarism. A correct academic version would paraphrase more independently and cite Yang and Clarke.

Question 21 — Employability communication

21(a) Matching the STAR categories



A STAR example is strongest when it moves from context -> responsibility -> what you did -> evidence of outcome.

STAR category	Matching statement	Why it fits
Situation	Statement 3	Provides the background context: a consultation on proposed legislative change.
Task	Statement 1	States the responsibility the person had within that situation.
Action	Statement 4	Explains the steps the person took.
Result	Statement 2	Reports the outcome and organisational benefit produced.

21(b) Classifying covering-letter statements

Statement type	Matching examples	Reasoning
1. Opening statements	F, L	They begin the letter by referring to the advertisement or vacancy source.
2. Experience and qualifications	A, B, C, G	They describe current role, duties, and relevant experience.
3. Reasons for applying	D, E	They explain motivation and fit for the job.
4. Closing statements	H, I, J, K	They end the letter politely and invite further contact or interview.

A competent covering letter usually follows this same structure: open clearly, prove suitability with evidence, explain motivation, then close professionally with availability and next-step language.

End note: this booklet solves every question visible in the uploaded Section B source. Section A was marked as unavailable in the source file.